Descriptive Written Test

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1, Smart Home Automation System:

A Smart home automation System is being designed to manage lighting, temperature, and Security based on user behaviour and environmental conditions. The System Must respond to real-time inputs Such as motion detection, temperature changes, and time.

Types:

- > Simple reflex agent
- > Model Based agent
- > Goal Based agent
- > utility Based agent

> Simple reflex Agent:

* Acts based on current inputs using of predefined rules.
* Examples: Switches on lights when motion is detected
* Advantages: Fast and Simple.

> Model Based Agents:

* Maintains an Internal State of the environment.

* Examples: Remember no occupancy & temperature changes
* Advantages: Fast and Simple.

Goal based agents:

* Select actions to achieve specific goals
* Eg: Maintain comfortable room temperature and Secures doors
* Advantages: Support planning and decision making.

Electricity based agents:

* Select action that maximize overall benefit.
* Eg: balance comfort, Security and electricity consumption.
* Advantage: compares alternatives to choose the best action.

Therefore:

The hybrid approach improves comfort energy efficiency, Security and personalization by combining immediate responses with planning by intelligent decision making.

2, A robot can use uninformed Search in algorithms to explore an unknown manage and find path to exit.

uniform cost Search (UCS):

> Insert Starting position.

> check whether it is the exist
> expand neighbouring position & updated cost
> repeat until exit found.

Advantage:

uses less memory and it optional when all steps cost are equal.

VCS	exponential in worst case	$O(b^d)$	potential on potential	$O(b^d)$
IOS				

Types Intelligent agents:

> Simple reflex: reacts to current of obstacles
> model based: Remembers explored path & usually
> Goal based: plans action to reasearch.
> utility based: consider softedly, energy & cost

Therefore:

VCS minimizes total path cost, while less reduces memory usage combining a model based agent with goal based planning allows explore maze

Algorithm	Time complexcibility	Space complexcibility
VCS	Exponential In worst case	Potential exponential
IOS	$O(b^d)$	$O(b^d)$

Types Intelligent Agents:

> Simple reflex: Reacts to current of obstacles
> model-based: Remembers explored paths & walls
> Goals-based: plans actions to research the exist
> utility-based: Consider Safety, energy & cost

Therefore:

VCS minimizes total path cost, while loss reduces memory usage combining a model based agent with goal based Planning allows explore maze.